

Vedlegg – Common maths / Vanlig matte

Scalar (dot) product:

$$\mathbf{a} \cdot \mathbf{b} \equiv \mathbf{b} \cdot \mathbf{a} = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} \cdot \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} = a_x b_x + a_y b_y + a_z b_z$$

Vector (cross) product:

$$\mathbf{a} \times \mathbf{b} \equiv -\mathbf{b} \times \mathbf{a} = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} \times \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} = \begin{bmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{bmatrix}$$

Vector norm and square-norm:

$$|\mathbf{a}| \equiv \|\mathbf{a}\| = \sqrt{a_x^2 + a_y^2 + a_z^2} \quad \mathbf{a}^2 \equiv \mathbf{a} \cdot \mathbf{a} \equiv |\mathbf{a}|^2 = a_x^2 + a_y^2 + a_z^2$$

Basis vectors:

$$\mathbf{e}_x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{e}_y = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{e}_z = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Dot- and cross-products of basis vectors:

$$\begin{array}{lll} \mathbf{e}_x \cdot \mathbf{e}_x = 1 & \mathbf{e}_x \cdot \mathbf{e}_y = 0 & \mathbf{e}_x \cdot \mathbf{e}_z = 0 \\ \mathbf{e}_y \cdot \mathbf{e}_x = 0 & \mathbf{e}_y \cdot \mathbf{e}_y = 1 & \mathbf{e}_y \cdot \mathbf{e}_z = 0 \\ \mathbf{e}_z \cdot \mathbf{e}_x = 0 & \mathbf{e}_z \cdot \mathbf{e}_y = 0 & \mathbf{e}_z \cdot \mathbf{e}_z = 1 \\ \\ \mathbf{e}_x \times \mathbf{e}_x = \mathbf{0} & \mathbf{e}_x \times \mathbf{e}_y = \mathbf{e}_z & \mathbf{e}_x \times \mathbf{e}_z = -\mathbf{e}_y \\ \mathbf{e}_y \times \mathbf{e}_x = -\mathbf{e}_z & \mathbf{e}_y \times \mathbf{e}_y = \mathbf{0} & \mathbf{e}_y \times \mathbf{e}_z = \mathbf{e}_x \\ \mathbf{e}_z \times \mathbf{e}_x = \mathbf{e}_y & \mathbf{e}_z \times \mathbf{e}_y = -\mathbf{e}_x & \mathbf{e}_z \times \mathbf{e}_z = \mathbf{0} \end{array}$$

Unit vector pointing from $\mathbf{r}_a$ to $\mathbf{r}_b$:

$$\frac{\mathbf{r}_b - \mathbf{r}_a}{|\mathbf{r}_b - \mathbf{r}_a|}$$

Derivatives and integrals

$$\frac{d \text{const.}}{dx} = 0 \quad \frac{dx^n}{dx} = n x^{n-1}$$

$$\int_a^b \text{const.} dx = (b - a) \cdot \text{const.}$$

$$\int_a^b x^n dx = \frac{1}{n+1} (b^{n+1} - a^{n+1}), \quad n \neq -1$$

$$\int_a^b x^{-1} dx = \ln(b/a)$$

Vedlegg (English) - List of common symbols

(Greek symbols)

κ	(kappa)	gravitational constant
μ	(mu)	viscosity coefficient
μ_k		coefficient of kinetic friction
μ_s		coefficient of static friction
Π	(capital pi)	entropy flux
ρ	(rho)	molar mass
Φ	(capital phi)	energy-mass flux
$\dot{\xi}$	(xi-dot)	rate of conversion
$\boldsymbol{\Omega}$	(capital omega)	angular-velocity matrix
$\boldsymbol{\omega}$	(omega)	angular-velocity vector

(Latin symbols)

A		usually: area; in some contexts: just a coefficient
$\mathcal{A}$	(script capital a)	energy-mass supply
$\mathcal{B}$	(script capital b)	magnetic flux
C		usually: molar heat capacity; in some contexts: just a coefficient
c		usually: speed of light; in some contexts: just a coefficient
E		energy-mass content
$\mathcal{E}$	(script capital e)	electric tension (line integral of electric field strength)
$\mathbf{F}$		momentum flux (surface force)
$\mathcal{F}$	(script capital f)	stress vector, traction.
$\mathbf{G}$		momentum supply (volume force)
$\mathbf{g}$		acceleration vector of free fall
h		surface coefficient of heat transfer
$\mathbf{I}_c$		moment of inertia (with respect to centre of mass-energy)
$\mathcal{I}$	(script capital i)	electric current
J		matter flux
k		elastic constant
$\mathbf{L}$		angular-momentum content
$\mathbf{M}$		angular-momentum flux (surface torque)
m		mass-energy content, typically at rest
N		matter content (amount of substance)
$\mathbf{P}$		momentum content
p		pressure
$\dot{Q}$	(q-dot)	heat flux
Q	(script capital q)	electric-charge content
R		usually: molar gas constant; in some contexts: radius
$\mathcal{R}$	(script capital r)	matter supply
$\mathbf{r}$		coordinate-position vector
S		entropy content
T		thermodynamic temperature (in kelvins)
$\mathcal{T}$	(script capital t)	angular-momentum supply (volume torque)
t		coordinate time
U		internal energy
V		volume
$\mathbf{v}$		coordinate-velocity vector
x		position coordinate, typically horizontal
y		position coordinate, typically horizontal
z		position coordinate, typically vertical

Vedlegg (English) – Balance laws

General form of balance law t is a *coordinate* time:

integral expression: $\text{vol. content}(t_1) = \text{vol. content}(t_0) + \int_{t_0}^{t_1} \text{influx}(t) \, dt + \int_{t_0}^{t_1} \text{supply}(t) \, dt$

differential expression: $\frac{d \text{ vol. content}(t)}{dt} = \text{influx}(t) + \text{supply}(t)$

The seven universal balance laws They apply to every physical phenomenon, in all physical theories:

	integral expression	differential expression
matter	$N(t_1) = N(t_0) + \int_{t_0}^{t_1} J(t) \, dt + \int_{t_0}^{t_1} \mathcal{R}(t) \, dt$	$\frac{dN(t)}{dt} = J(t) + \mathcal{R}(t)$
momentum	$\mathbf{P}(t_1) = \mathbf{P}(t_0) + \int_{t_0}^{t_1} \mathbf{F}(t) \, dt + \int_{t_0}^{t_1} \mathbf{G}(t) \, dt$	$\frac{d\mathbf{P}(t)}{dt} = \mathbf{F}(t) + \mathbf{G}(t)$
energy	$E(t_1) = E(t_0) + \int_{t_0}^{t_1} \Phi(t) \, dt + \int_{t_0}^{t_1} \mathcal{A}(t) \, dt$	$\frac{dE(t)}{dt} = \Phi(t) + \mathcal{A}(t)$
angular momentum	$\mathbf{L}(t_1) = \mathbf{L}(t_0) + \int_{t_0}^{t_1} \mathbf{M}(t) \, dt + \int_{t_0}^{t_1} \mathcal{T}(t) \, dt$	$\frac{d\mathbf{L}(t)}{dt} = \mathbf{M}(t) + \mathcal{T}(t)$
entropy	$S(t_1) \geq S(t_0) + \int_{t_0}^{t_1} \Pi(t) \, dt$	$\frac{dS(t)}{dt} \geq \Pi(t)$
electric charge	$Q(t_1) = Q(t_0) + \int_{t_0}^{t_1} I(t) \, dt$	$\frac{dQ(t)}{dt} = I(t)$
magnetic flux	$\mathcal{B}(t_1) = \mathcal{B}(t_0) - \int_{t_0}^{t_1} \mathcal{E}(t) \, dt$	$\frac{d\mathcal{B}(t)}{dt} = -\mathcal{E}(t)$

Kinematic relations

$$\frac{d\mathbf{r}(t)}{dt} = \mathbf{v}(t)$$

$$\frac{d\mathbf{v}(t)}{dt} = \mathbf{a}(t)$$

$$\frac{d\mathbf{I}_c(t)}{dt} = \boldsymbol{\Omega}(t) \mathbf{I}_c(t) - \mathbf{I}_c(t) \boldsymbol{\Omega}(t)$$

Vedlegg (English) - Some constitutive relations Most of these apply only at speeds lower than light's, weak gravity, and in the absence of electromagnetic fields:

Name	formula	notes & requirements
Mass-energy for matter	$m = \rho N$	ρ approx. same throughout control volume
Newton's formula for momentum of matter	$\mathbf{P} = m \mathbf{v}$	small control volume; $\mathbf{v}$: velocity of matter
... for rigid body	$\mathbf{P} = m \mathbf{v}_c$	control vol. of any size, but rigid motion; $\mathbf{v}_c$: velocity of mass centre
Gravitational momentum supply (volume force)	$\mathbf{G} = -m g \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	coords at rest with ground; z : vertical; $g \approx 9.8 \text{ N/kg}$
Newton's law of gravitation (volume force)	$\mathbf{G}_a = \kappa \frac{m_a m_b}{ \mathbf{r}_a - \mathbf{r}_b ^3} (\mathbf{r}_b - \mathbf{r}_a)$	$\kappa \approx 6.674 \times 10^{-11} \text{ N m}^2/\text{kg}^2$
Hooke's law for surface force / momentum efflux	$\mathbf{F}_{\rightsquigarrow a} = -k (\mathbf{r}_a - \mathbf{r}_b - l_n) \frac{\mathbf{r}_a - \mathbf{r}_b}{ \mathbf{r}_a - \mathbf{r}_b }$	
Non-Hookean law, rubber band	$\mathbf{F}_{\rightsquigarrow a} = \begin{cases} 0 \text{ N} & \text{if } \mathbf{r}_a - \mathbf{r}_b \leq l_n \\ -k (\mathbf{r}_a - \mathbf{r}_b - l_n) \frac{\mathbf{r}_a - \mathbf{r}_b}{ \mathbf{r}_a - \mathbf{r}_b } & \text{if } \mathbf{r}_a - \mathbf{r}_b > l_n \end{cases}$	
Kinetic friction	$\mathbf{F}_k = -\frac{\mathbf{v}}{ \mathbf{v} } \mu_k F_n$	F_n : normal force
Energy content for matter	$E = U + E_k + E_p$	
Kinetic energy	$E_k = \frac{1}{2} m \mathbf{v}^2$	small control volume
... for rigid body	$E_k = \frac{1}{2} m \mathbf{v}_c^2 + \frac{1}{2} \boldsymbol{\omega} \cdot \mathbf{I}_c \boldsymbol{\omega}$	$\mathbf{v}_c$: velocity of mass centre
Gravitational potential energy	$E_p = m g z$	control volume of small vertical dimension; z : vertical
... for rigid body	$E_p = m g z_c$	z_c : height of mass-centre
Energy flux for matter	$\Phi = \dot{Q} + \mathbf{F} \cdot \mathbf{v}$	no matter flux; $\mathbf{v}$: matter velocity at control surface (same over whole surface)
Energy supply	$\mathcal{A} \approx 0 \text{ J/s}$	small cosmological scales, inertial coordinates
Ideal-gas law, also under compression or expansion	$p := \frac{F}{A} = \frac{RNT}{V} - \mu \frac{1}{V} \frac{dV}{dt}$	homogeneous conditions
Internal energy of ideal gas	$U = CNT$	T approx. same throughout control volume
Newton's law of cooling	$\dot{Q}_{a \rightsquigarrow b} = A h (T_a - T_b)$	T_a, T_b approx. same over whole control surface
Angular-momentum content for matter	$\mathbf{L} = \mathbf{r} \times m \mathbf{v}$	small control volume (so $\mathbf{v}$ approx. same throughout)
... for rigid body	$\mathbf{L} = \mathbf{r}_c \times m \mathbf{v}_c + \mathbf{I}_c \boldsymbol{\omega}$	$\mathbf{r}_c$: position of mass centre
Volume torque / angular-momentum supply	$\mathcal{T} = \mathbf{r}_c \times \mathbf{G}$	$\mathbf{r}_c$: position of mass centre
Surface torque / angular-momentum flux	$\mathbf{M} = \mathbf{r} \times \mathbf{F}$	surf. force / momentum flux $\mathbf{F}$ concentrated at position $\mathbf{r}$

Main units

t	$\mathbf{r}$	N	$\mathbf{P}$	E	$\mathbf{L}$	Q	$\mathcal{B}$	S	T
s	m	mol	N · s	J	N · m · s	C	Wb	J/K	K

$$\text{flux units} = \text{supply units} = \frac{\text{quantity units}}{\text{time units}}$$

$$J = N \cdot m = \text{kg} \cdot \text{m}^2/\text{s}^2$$

Important physical principles

■ **Extensivity.** Applies to the contents, fluxes, and supplies of all seven main quantities:

Contents: if we divide a volume into (non-overlapping) sub-volumes, then the net content of a quantity in the full volume is equal to the sum of the contents in all sub-volumes.

Fluxes: if we divide a surface into (non-overlapping) partial surfaces, then the net flux of a quantity through the whole surface is equal to the sum of the fluxes through all partial surfaces. (The crossing directions must match!)

Supplies: if we divide a volume into (non-overlapping) sub-volumes, then the net supply of a quantity in the full volume is equal to the sum of the supplies in all sub-volumes.

■ **Symmetry of fluxes.** The flux of a quantity through a surface in one direction (say, from side A to side B) is equal to *minus* the flux in the other direction (from side B to side A).

Applies to the fluxes of all seven main quantities.

In the case of momentum flux (surface force) this principle is called *Newton's third law* or the *law of action and reaction*.

❗ The principle applies only to the whole flux, not to the terms (mechanical, thermal, etc) which may compose it. For example, for energy flux given by $\dot{Q} + \mathbf{F} \cdot \mathbf{v}$ we have

$$\dot{Q}_{\rightsquigarrow B} + \mathbf{F}_{A \rightsquigarrow B} \cdot \mathbf{v}_A = -(\dot{Q}_{\rightsquigarrow A} + \mathbf{F}_{B \rightsquigarrow A} \cdot \mathbf{v}_B)$$

but it may happen that $\dot{Q}_{\rightsquigarrow B} \neq -\dot{Q}_{\rightsquigarrow A}$ and $\mathbf{v}_A \neq \mathbf{v}_B$.

Alternative names

Momentum flux	≡	Surface force	≡	Contact force
Momentum supply	≡	Volume force	≡	Body force
Angular-momentum flux	≡	Surface torque	≡	Contact torque
Angular-momentum supply	≡	Volume torque	≡	Body torque

Vedlegg (bokmål) - Liste over vanlige symboler

(Greske symboler)

κ	(kappa)	gravitasjonskonstant
μ	(mu)	viskositetskoeffisient
μ_k		koeffisient for kinetisk friksjon
μ_s		koeffisient for statisk friksjon
Π	(capital pi)	entropifluks
ρ	(rho)	molar masse
Φ	(capital phi)	energi-fluks
Ω	(capital omega)	rotasjonshastighetsmatrise
ω	(omega)	rotasjonshastighetsvektor

(Latinske symboler)

A		vanligvis: areal; i noen sammenhenger: bare en koeffisient
$\mathcal{A}$	(script capital a)	energi-kilde
$\mathcal{B}$	(script capital b)	magnetisk fluks
C		vanligvis: molar varmekapasitet; i noen sammenhenger: bare en koeffisient
c		vanligvis: lysets hastighet; i noen sammenhenger: bare en koeffisient
E		energi-innhold
$\mathcal{E}$	(script capital e)	elektrisk potensialforskjell (elektrisk spenning)
$\mathbf{F}$		momentumsfluks (overflatekraft)
$\mathcal{F}$	(script capital f)	spenningsvektor
$\mathbf{G}$		momentumskilde (volumkraft)
$\mathbf{g}$		akselerasjonsvektor for fritt fall
h		overflatekoeffisient for varmeoverføring
$\mathbf{I}_c$		treghetsmoment (med hensyn til sentrum av masse)
$\mathcal{I}$	(script capital i)	elektrisk strøm
J		materiefluks
k		elastisk konstant
$\mathbf{L}$		dreieimpulsinnhold (dreieimpuls i kontrollvolum)
$\mathbf{M}$		dreieimpulssfluks (dreiemoment/torque ved overflaten)
m		masse-innhold, typisk i ro
N		materieinnhold (mengde substans i kontrollvolumet)
$\mathbf{P}$		momentumsinnhold
p		trykk
$\dot{Q}$	(q-dot)	varmefluks
Q	(script capital q)	mengde elektrisk ladning
R		vanligvis: molar gasskonstant; i noen sammenhenger: radius
$\mathcal{R}$	(script capital r)	materiekilde
$\mathbf{r}$		koordinat-posisjonsvektor
S		entropiinnhold
T		termodynamisk temperatur (i kelvin)
$\mathcal{T}$	(script capital t)	dreieimpulskilde (dreiemoment/torque på volumet)
t		tidskoordinat
U		indre energi
V		volum
$\mathbf{v}$		hastighetsvektor (koordinat)
x		posisjon (koordinat), typisk horisontal
y		posisjon (koordinat), typisk horisontal
z		posisjon (koordinat), typisk vertikal

Vedlegg (bokmål) – Balanselovene

Generell form på en balanselov t er her en *tidskoordinat* ikke et *tidspunkt*.

integralform: $\text{vol. innhold}(t_1) = \text{vol. innhold}(t_0) + \int_{t_0}^{t_1} \text{innfluks}(t) \, dt + \int_{t_0}^{t_1} \text{kilde}(t) \, dt$

differensialform: $\frac{d \text{ vol. innhold}(t)}{dt} = \text{innfluks}(t) + \text{kilde}(t)$

De syv universelle balanselovene De gjelder for alle fysiske fenomen, alle fysiske teorier:

	integralform	differensialform
materie	$N(t_1) = N(t_0) + \int_{t_0}^{t_1} J(t) \, dt + \int_{t_0}^{t_1} \mathcal{R}(t) \, dt$	$\frac{dN(t)}{dt} = J(t) + \mathcal{R}(t)$
momentum	$\mathbf{P}(t_1) = \mathbf{P}(t_0) + \int_{t_0}^{t_1} \mathbf{F}(t) \, dt + \int_{t_0}^{t_1} \mathbf{G}(t) \, dt$	$\frac{d\mathbf{P}(t)}{dt} = \mathbf{F}(t) + \mathbf{G}(t)$
energi	$E(t_1) = E(t_0) + \int_{t_0}^{t_1} \Phi(t) \, dt + \int_{t_0}^{t_1} \mathcal{A}(t) \, dt$	$\frac{dE(t)}{dt} = \Phi(t) + \mathcal{A}(t)$
dreieimpuls	$\mathbf{L}(t_1) = \mathbf{L}(t_0) + \int_{t_0}^{t_1} \mathbf{M}(t) \, dt + \int_{t_0}^{t_1} \mathcal{T}(t) \, dt$	$\frac{d\mathbf{L}(t)}{dt} = \mathbf{M}(t) + \mathcal{T}(t)$
entropi	$S(t_1) \geq S(t_0) + \int_{t_0}^{t_1} \Pi(t) \, dt$	$\frac{dS(t)}{dt} \geq \Pi(t)$
elektrisk ladning	$Q(t_1) = Q(t_0) + \int_{t_0}^{t_1} I(t) \, dt$	$\frac{dQ(t)}{dt} = I(t)$
magnetisk fluks	$\mathcal{B}(t_1) = \mathcal{B}(t_0) - \int_{t_0}^{t_1} \mathcal{E}(t) \, dt$	$\frac{d\mathcal{B}(t)}{dt} = -\mathcal{E}(t)$

Kjært barn har mange navn. Momentum kalles også *bevegelsesmengde*, dreieimpuls kalles også *rotasjonsmengde*, og materie kalles ofte *stoffmengde*.

Kinematikk

$$\frac{d\mathbf{r}(t)}{dt} = \mathbf{v}(t)$$

$$\frac{d\mathbf{v}(t)}{dt} = \mathbf{a}(t)$$

$$\frac{d\mathbf{I}_c(t)}{dt} = \boldsymbol{\Omega}(t) \mathbf{I}_c(t) - \mathbf{I}_c(t) \boldsymbol{\Omega}(t)$$

Vedlegg (bokmål) - Noen konstitutive relasjoner De fleste av disse gjelder bare ved lave hastigheter enn lysets, svak gravitasjon, og når det ikke finnes elektromagnetiske felt:

<i>Navn</i>	<i>formel</i>	<i>notater & krav</i>
Masse for materie	$m = \rho N$	ρ omtrent samme over hele kontrollvolumet
Newtons formel for momentum til materie	$\mathbf{P} = m \mathbf{v}$	punktlignende kontrollvolum; $\mathbf{v}$: materies hastighet
... for stivt legeme	$\mathbf{P} = m \mathbf{v}_c$	$\mathbf{v}_c$: hastighet til massesenter
Tyngdekraft som en momentumkilde (volumkraft)	$\mathbf{G} = -m \mathbf{g} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	koordinater i ro relativt til bakken; z: vertikal; $g \approx 9.8 \text{ N/kg}$
Newtons lov for tyngdekraften (volumkraft)	$\mathbf{G}_a = \kappa \frac{m_a m_b}{ \mathbf{r}_a - \mathbf{r}_b ^3} (\mathbf{r}_b - \mathbf{r}_a)$	$\kappa \approx 6.674 \times 10^{-11} \text{ N m}^2/\text{kg}^2$
Hookes lov for overflatekraft / momentumsutfluks	$\mathbf{F}_{\rightsquigarrow a} = -k (\mathbf{r}_a - \mathbf{r}_b - l_n) \frac{\mathbf{r}_a - \mathbf{r}_b}{ \mathbf{r}_a - \mathbf{r}_b }$	
ikke-Hookes fjær, gummitrikk	$\mathbf{F}_{\rightsquigarrow a} = \begin{cases} 0 \text{ N} \\ -k (\mathbf{r}_a - \mathbf{r}_b - l_n) \frac{\mathbf{r}_a - \mathbf{r}_b}{ \mathbf{r}_a - \mathbf{r}_b } \end{cases}$	hvis $ \mathbf{r}_a - \mathbf{r}_b \leq l_n$ hvis $ \mathbf{r}_a - \mathbf{r}_b > l_n$
Kinetisk friksjon	$\mathbf{F}_k = -\frac{\mathbf{v}}{ \mathbf{v} } \mu_k F_n$	F_n : normalkraft
Energiinnhold for materie	$E = U + E_k + E_p$	
Kinetisk energi	$E_k = \frac{1}{2} m \mathbf{v}^2$	punktlignende kontrollvolum
... for stivt legeme	$E_k = \frac{1}{2} m \mathbf{v}_c^2 + \frac{1}{2} \boldsymbol{\omega} \cdot \mathbf{I}_c \boldsymbol{\omega}$	$\mathbf{v}_c$: hastighet til massesenter
Gravit. potensiell energi	$E_p = m g z$	kontrollvolum med liten vertikal dimensjon; z: vertikal
... for stivt legeme	$E_k = m g z_c$	z_c : høyde til massesenter
Energifluks for materie	$\Phi = \dot{Q} + \mathbf{F} \cdot \mathbf{v}$	ingen materiefuks; $\mathbf{v}$: materiehastighet på kontrollflaten (samme over hele kontrollflaten)
Energikilde	$\mathcal{A} \approx 0 \text{ J/s}$	små kosmologiske skalaer, inertielle koordinater
Idealgassloven, også ved kompresjon eller ekspansjon	$p := \frac{F}{A} = \frac{RNT}{V} - \mu \frac{1}{V} \frac{dV}{dt}$	homogene forhold
Indre energi for idealgass	$U = CNT$	T omtrent samme gjennom kontrollvolumet
Newtons kjølelov	$\dot{Q}_{a \rightsquigarrow b} = A h (T_a - T_b)$	T_a, T_b omtrent samme over hele kontrollflaten
Dreieimpuls for stivt legeme	$\mathbf{L} = \mathbf{r}_c \times m \mathbf{v}_c + \mathbf{I}_c \boldsymbol{\omega}$	$\mathbf{r}_c$: posisjon for massesenter
Volum-torque / Dreieimpulskilde	$\mathcal{T} = \mathbf{r}_c \times \mathbf{G}$	$\mathbf{r}_c$: posisjon for massesenter
Overflate-torque / Dreieimpulssfluks	$\mathbf{M} = \mathbf{r} \times \mathbf{F}$	momentumsfluks / overflatekraft $\mathbf{F}$ konsentrert ved posisjon $\mathbf{r}$

Hovedenheter

t	r	N	P	E	L	Q	$\mathcal{B}$	S	T
s	m	mol	N · s	J	N · m · s	C	Wb	J/K	K

$$\text{fluksenhet} = \text{kildeenhet} = \frac{\text{størrelsesenhet}}{\text{tidsenhet}}$$

$$J = N \cdot m = \text{kg} \cdot \text{m}^2/\text{s}^2$$

Viktige fysikkprinsipper

■ **Ekstensivitet.** Gjelder for mengder, flukser, og kilder for alle syv hovedstørrelsene:

Mengder: Om vi deler et volum inn i (ikke-overlappende) del-volum så er totalinnholdet av en størrelse i hele volumet lik summen av størrelsene i del-volumene.

Fluks: Om vi deler en flate inn i (ikke-overlappende) del-flater så er samlet fluks av en størrelse gjennom hele flaten lik summen av fluks gjennom alle del-flatene. (Krysningsretningene må stemme overens!)

Kilder: Om vi deler et volum inn i (ikke-overlappende) del-volum så er den samlede kilden av en størrelse i hele volumet lik summen av kildene i alle del-volum

■ **Flukssymmetri:** Fluks av en størrelse gjennom en flate i en retning (fra side A til side B) er lik negativ fluks i den andre retningen (fra side B til side A).

Gjelder fluks for alle de syv hovedstørrelsene.

Når det gjelder momentumfluks (overflatekraft) kalles dette prinsippet *Newtons tredje lov*, eller *lov om kraft og motkraft*

❗ Prinsippet gjelder kun for hele fluksen, og ikke de ulike delene (mekanisk, termisk osv.) som den stammer fra. For eksempel for energifluks gitt som: $\dot{Q} + \mathbf{F} \cdot \mathbf{v}$ så har vi:

$$\dot{Q}_{\rightsquigarrow B} + \mathbf{F}_{A \rightsquigarrow B} \cdot \mathbf{v}_A = -(\dot{Q}_{\rightsquigarrow A} + \mathbf{F}_{B \rightsquigarrow A} \cdot \mathbf{v}_B)$$

men det kan være at $\dot{Q}_{\rightsquigarrow B} \neq -\dot{Q}_{\rightsquigarrow A}$ og $\mathbf{v}_A \neq \mathbf{v}_B$.

Alternative navn

Momentumfluks	≡	Overflatekraft	≡	Kontaktkraft
Momentumkilde	≡	Volumkraft	≡	Kroppskraft
Angulærmomentum-fluks	≡	Overflatemoment	≡	Kraftmoment
Angulærmomentum-kilde	≡	Volummoment	≡	Kroppsmoment